COMPUTER SCIENCE –LEVEL 6
UNDERSTANDING COMPUTER ORGANIZATION AND ARCHITECTURE I
CT/CU/CS/CR/01/6/8
1 1/2 HOURS
AUGUST 2025

JKUAT TVET INSTITUTE – KAREN CAMPUS

WRITTEN ASSESSMENT

CAT 2

Attempt all the questions.

SECTION A (20 marks)

1. Define the following terms as related to computer organisation
   a. Volatile memory
   b. Computer organization (4 marks)
2. PC motherboards have buses for expansion and external devices. list **any three** types of buses (3 marks)
3. Give any two differences between Computer Architecture and Computer Organization? (4 marks)
4. Computer systems utilise different types of RAM for temporary data storage, each with distinct features and operations. Differentiate between Static RAM (SRAM) and Dynamic RAM(DRAM). (4 Marks)
5. Output devices are essential components in a computer system as they allow users to receive information from the computer in a human-readable form. List any five output devices. (5 marks

SECTION B (30 marks)

6. The Central Processing Unit has its main function as the brain of the computer.
   a. Describe the fetch, decode, and execute cycle of the instruction representation (6 marks)
   b. List any four functions of the control unit (4 marks)
7. Secondary storage technologies are used for retaining data permanently. These storage options vary in terms of speed, capacity, cost, and usage. Discuss any FIVE secondary storage technologies (10 marks)

8. .

a. You are planning to buy on your personal computer to start a cybercafe business. List **any four** key specification factors that should be considered when selecting input/output for your computer (4 marks)

b. Your computer's performance heavily relies on how efficiently memory is managed. Explain **any three types of computer memory** (6 marks)

CAT 2

4.5/8

JKUAT CONTINOUS ASSESSEMENT TESTS

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REG NO: TOC113-0032-0038/2024

COURSE: Computer Science YEAR OF STUDY: 1

SHEET NO:

UNIT CODE: TITLE: Computer Architecture & Organization

DATE: 7/8/2025

NOTE: This stationery will be used for Continuous Assessment work only.

It will be a breach of examination regulations to use it otherwise.

1a) Volatile memory refers to the memory which looses it data even when the power is switched off (1)

b) Computer Organization refers to the interconnection of hardware components to perform various functions (2)

2. i) Data bus (1)

ii) Address bus (1)

iii) Control bus (1)

3. i) Computer Architecture is concerned with ~~refers to~~ the (1) internal working of components in a computer system while computer organization is concerned ~~refers to~~ the structural and behaviour of the computer.

ii) Computer Architecture requires high level design while computer organisation requires low-level design.

4. i) SRAM is faster while (1) DRAM (1) slow.

ii) SRAM stores data using flip flop while (1) DRAM uses capacitors to store data.

iii) SRAM is expensive while DRAM is cheap.

iv) SRAM does not require refreshing while DRAM requires refreshing.

5) i) Printer (1)

ii) Monitor (1)

iii) Speakers (1)

iv) Touchscreen (1)

v) Projector (1)

SECTION B

6a) Fetch Stage

→ The CPU receives data and instructions from the main memory and stores them in the memory location which is known as the register

Decode Stage

- After the data and instructions is stored in the register it is then brought in the decode stage for interpretation.

Execution Stage

→ After data and information ~~the~~ is taken to the decode stage for interpretation, it is then taken to be executed and processed by the CPU then after the ~~executed~~ execution, the cycle execution process starts again and continues.

b) i) It ensures instructions are runned smoothly

ii) It is the manager for telling all components what to do

iii) It ensures all units are arranged in the correct sequential ~~sequential~~ order

iv) It ~~acts~~ interprets the instruction to binary language.

7. i) HDD - Hard Disk Drive is a non-volatile memory ~~storage~~ which is crucial for storing data ~~for future reference~~ and it's usually larger than the SSD and less resistant to shock

ii) SSD - Solid State Drive is a non-volatile memory which is also crucial for storing data permanently and it is small in size and compact to be used in devices like laptops and other electronic devices.

iii) Optical Disks → Optical disks like Blue Ray Disk and DVD (Digital Versatile Disk) are also considered to store data permanently for a specified period of time in terms of their durability

iv) Flash Memory - The Flash memory is usually considered non-volatile because of its durability to store data for a long period of time and its also portable in devices like laptops and smartphones.

v) Zip Disks - Zip disks is a storage technology devices which is removable and considered non-volatile storage device due to its durability to store data

8a) i) Speed (1)

ii) Durability (1)

iii) Running cost and Initial cost (1)

iv) Portability and Upgradability (1)

~~b) Cache mem~~

b) i) Cache memory - This is a small and compact memory which is crucial for ~~storing~~ temporary storing frequently used data. (2)

ii) Volatile memory - This is a memory crucial for storing data temporarily and data can be easily be lost when the power is switched off. (1)

iii) Non Volatile memory - This is a memory crucial for storing data permanently and data can be retained even when the power is switched off. (1)